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A Comprehensive Review on Machine Learning and Data-Driven Approaches for Song Popularity Prediction

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1. INTRODUCTION

- Music industry digitalization has transformed how songs are created, distributed, and consumed.
- Streaming platforms like Spotify,YouTube, and Apple Music dominate music discovery.
- Predicting a song’s popularity helps artists, producers, and record labels plan releases.
- Popularity depends on acoustic, lyrical, and social factors.
- Machine Learning (ML) enables proactive prediction from large-scale datasets.

2.PROBLEM STATEMENT

Complexity of Song Popularity Prediction

- Depends on acoustic, lyrical & social factors
- Affected by listener behaviour and cultural trends
- Multifactorial – no single feature ensures success

Challenges:

- Inconsistent datasets & metrics
- English-centric data bias
- Poor interpretability of ML models

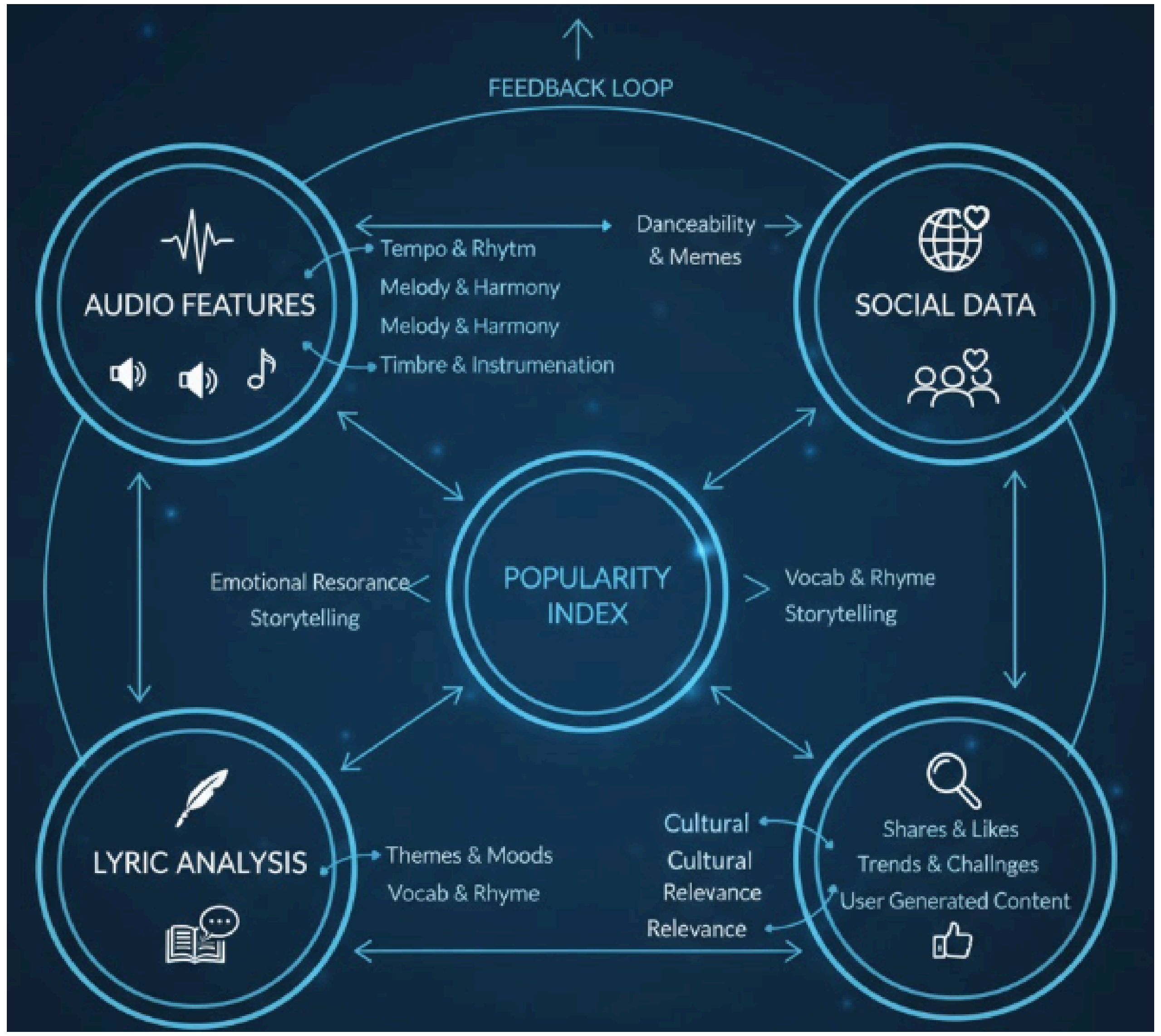

Fig 1:How audio + lyrics + social data interact to influence popularity.

3. THEORIES

Traditional Models:

- Support Vector Machine (SVM), Decision Tree, Random Forest.

Advanced Models:

- XGBoost, LightGBM for ensemble learning.
- BERT for lyric sentiment and emotion analysis.

Multimodal Learning:

- Combines audio, text, and metadata for better accuracy.

Core Challenges:

- Data imbalance across genres.
- Limited interpretability and explainability.
- Regional diversity and dataset availability.

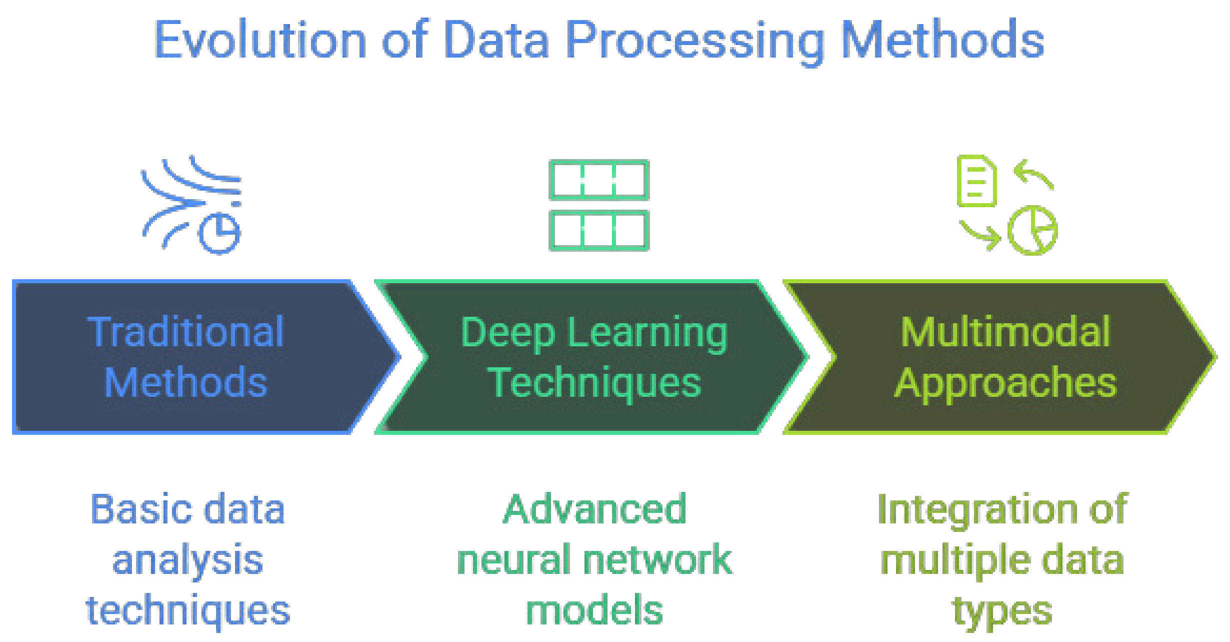

Fig 2: Evaluation of Data Processing Methods

4. METHODS

Study (Year)	Model / Technique	Data Source	Best Performance / Findings
Ramesh et al. (2021)	Random Forest, SVM, Decision Tree	Spotify (Metadata + Lyrics)	85.06% accuracy – Random Forest best
Wawo & Prasetyo (2023)	BERT (NLP)	Lyrics Dataset (52K songs)	87% accuracy – strong sentiment detection
Bhatia et al. (2022)	ML vs. Deep Learning (NN)	Spotify Streaming Data	DL slightly better – energy, loudness key
Ge et al. (2020)	PCA + Model Blending (RF + KNN + DT)	Custom Dataset	MSE = 4.96 – improved robustness
Lee & Hashim (2022)	Random Forest, XGBoost	Spotify + YouTube	79.6% accuracy – multimodal data improved results

5. RESULTS & DESCRIPTIVE STATISTICS

Overview :

- Reviewed 10 research papers (2020–2025)
- Covered ML, Deep Learning, NLP, Hybrid & Regional models
- Datasets: 7K – 1.8M songs (Spotify, YouTube, Billboard)
- Common features: Energy, Loudness, Tempo, Valence, Sentiment, Engagement

Performance Summary :

- Random Forest / XGBoost / LightGBM → Accuracy >80%
- BERT (Lyrics) → 87% Accuracy
- MLP (Hindi) → F1 = 0.88
- Hybrid (PCA + RF + KNN) → MSE ≈ 4.96

6. HYPOTHESES AND SCIENTIFIC DISCUSSION

1 - Ensemble models outperform single/deep models.

- Proof: Highest accuracy & stability across datasets.
- Practical Use: Easy to deploy, interpretable for streaming platforms.

2 - Multimodal integration (audio + lyrics + social) boosts accuracy.

- Proof: Fusion models improved accuracy by 10–15%.
- Practical Use: Helps music platforms personalize recommendations.

3 - Lyrical sentiment strongly influences popularity.

- Proof: BERT model → 87% accuracy using lyrics sentiment.
- Practical Use: Artists can assess lyrical emotion before release.

4 - Regional datasets enable localized predictions.

- Proof: Native Language datasets successful (F1 = 0.88).
- Practical Use: Supports regional music analytics and local industry growth.

7. CONCLUSIONS, SIGNIFICANCE

Conclusion :

- Ensemble & hybrid models = most consistent and reliable.
- Deep learning adds lyric & sentiment depth but needs large data.
- Regional studies prove ML adaptability across languages.

Significance

- Promotes data-driven creativity in the music industry.
- Builds a foundation for explainable, cross-platform prediction systems.
- Encourages inclusion of regional & multilingual content.

8. FUTURE SCOPE

- Create open multimodal datasets (audio, lyrics, social data).
- Add regional & multilingual tracks for global diversity.
- Use Explainable AI (SHAP, LIME) for transparency.
- Apply temporal models to track popularity trends over time.
- Build cross-platform frameworks for unified prediction.
- Enable Human–AI tools to guide artists during creation

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